

4.6.5

EE25BTECH11017 - Mahesh Chollangi

Question:

Find the equation of the line passing through $(2, 1, -1)$ and parallel to the line $\mathbf{r} = (\mathbf{i} + \mathbf{j}) + \lambda (2\mathbf{i} - \mathbf{j} + \mathbf{k})$. Also, find the distance between these two lines.

Solution:

Let the direction vector of the given line be

$$\mathbf{m} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \quad (0.1)$$

The point through which the required line passes is

$$\mathbf{A} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \quad (0.2)$$

Therefore, the equation of the required line is

$$\mathbf{r}_1 = \mathbf{A} + \lambda \mathbf{m} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \quad (0.3)$$

The equation of the given line can be written as

$$\mathbf{r}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \quad (0.4)$$

The distance d between two skew lines $\mathbf{r}_1 = \mathbf{a}_1 + \lambda \mathbf{m}$ and $\mathbf{r}_2 = \mathbf{a}_2 + \mu \mathbf{m}$ (with the same direction vector) is given by

$$d = \frac{\|(\mathbf{A} - \mathbf{B}) \times \mathbf{m}\|}{\|\mathbf{m}\|} \quad (0.5)$$

where $\mathbf{B} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

Compute $\mathbf{A} - \mathbf{B}$:

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \quad (0.6)$$

Now, compute the cross product:

$$(\mathbf{A} - \mathbf{B}) \times \mathbf{m} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & -1 \\ 2 & -1 & 1 \end{vmatrix} \quad (0.7)$$

Expanding this determinant,

$$= \mathbf{i}(0 \times 1 - (-1) \times (-1)) - \mathbf{j}(1 \times 1 - (-1) \times 2) + \mathbf{k}(1 \times (-1) - 0 \times 2) \quad (0.8)$$

$$= \mathbf{i}(0 - 1) - \mathbf{j}(1 + 2) + \mathbf{k}(-1 - 0) \quad (0.9)$$

$$= -\mathbf{i} - 3\mathbf{j} - \mathbf{k} \quad (0.10)$$

Therefore,

$$\|(\mathbf{A} - \mathbf{B}) \times \mathbf{m}\| = \sqrt{1^2 + 3^2 + 1^2} = \sqrt{11} \quad (0.11)$$

Now, compute $\|\mathbf{m}\|$:

$$\|\mathbf{m}\| = \sqrt{2^2 + (-1)^2 + 1^2} = \sqrt{4 + 1 + 1} = \sqrt{6} \quad (0.12)$$

Thus, the required distance is:

$$d = \frac{\sqrt{11}}{\sqrt{6}} \quad (0.13)$$

Therefore, the equation of the required line is:

$$\mathbf{r}_1 = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \quad (0.14)$$

and the distance between the two lines is

$$d = \frac{\sqrt{11}}{\sqrt{6}} \quad (0.15)$$

Distance between lines = -2003689232.0000

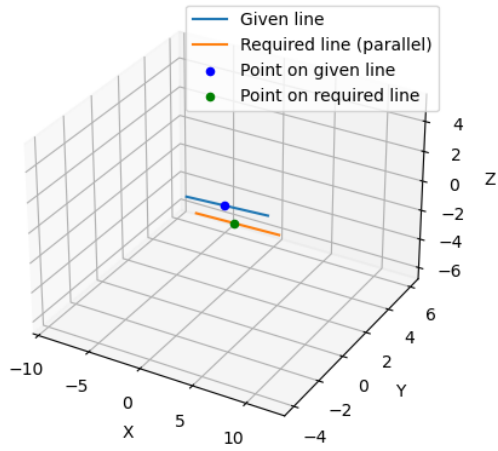


Fig. 0.1: 3D Plot